

Immunology-inspired Swarm Coordination with Application to Multiple Combat UGV Systems

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Abstract—Immune system has several interesting and desirable properties such as adaptability, robustness, flexibility, archival memory, and distributed cognition abilities. This paper presents a multi-agent scheme with swarm coordination functions inspired by immune system. It is shown that by using local communication, each agent is able to select its favorite strategy based on distributed sensing, which leads to successful swarm behavior, and performs well even under dynamically changing environment. Furthermore, in the proposed scheme a thymus-like global coordination mechanism called a critic here is introduced to prevent undesirable crowding, thus improving the overall swarming behavior as verified by numerical simulation on a collection of Unmanned Ground Vehicles (UGVs).

Index Terms – Swarm Cooperation, Bio-inspired, Cooperative Operation

I. INTRODUCTION

Swarm coordination is a concept representing the idea of distributing the functionality of a large agent among smaller, less-expensive and cooperative agents. A successful swarm coordination algorithm exhibits several advantages. It reduces cost, enhances reconfigurability and makes the overall system robust. It bears drastic potentials in the application of scientific and especially military purposes. However, designing an efficient swarm coordination scheme continues to be a challenge. As automatic systems become more complicated and system scales become larger, traditional swarm coordination methods encounter even greater difficulties. Because of these difficulties, engineers turn to natural biological systems for inspiration. Biological systems are highly intelligent systems. For instance, a flock of wild geese can deploy in a “V” formation to minimize wind resistance. Similarly, trillions of human antibodies can self-organize to mount a proper and adequate response to invading antigens. These phenomena intrigue people by their abilities to cope with the unpredictable natural environment.

For the last few years, a number of papers have been published regarding immunological-inspired topics. De Castro and Von Zuben proposed a clone selection algorithm named CLONALG [1] and implemented it successfully in pattern identification and optimization of multimodal function. After that, various immunological algorithms have been proposed to solve problems such as multi-objective optimization [2], [3] and combination optimization [4].

However, most of the proposed methods are either in a pure theoretical stage, or working on function peak value(s) problems etc., only few have really dealt with practical engineering problems [5-9].

It is noted that many existing swarm coordination schemes for multi-agent systems are based on a centralized operation. For example, Albus and Stentz [5] and Schenato, L. along with his colleagues [6] have both based their results on a centralized, hierarchical decision-making approach; which limits its application due to large computational and communications burden. Some other attempt on behavioral artificial intelligence (AI) methods have also been made but bottlenecks remain when coping with a complex and changing environment [7]. For dynamic environments having a relatively slow rate of change, an alternative approach for coordination is to repeatedly apply a method with guaranteed coverage [8]. However, this approach requires essentially that the frequency of environmental change is less than the bandwidth or refresh frequency.

In this paper, based upon the observation on the immune system (a remarkable example of highly scalable distributed system [9]), we proposed a new approach for cooperative swarm coordination of multiple combat UGVs. In this method, each agent in the system perceives its environment and its neighbors' behaviors, determining its own behavior independently, and then cooperates with other agents to achieve the global optimization, which is similar to the behavior of the immune system. The inner communicates are made through a series of local, peer to peer and implicit information exchange.

II. FOUNDAMENTALS OF IMMUNE SYSTEM

A. Overview

The protection system that eliminates foreign substances that invade a living body is called the immune system [10].

The human immune system works on two levels with the general goal of pathogen control: a general response mechanism, called innate immunity that does not directly respond to any specific pathogen and a specific, antibody-mediated response mechanism called acquired immunity.

B. B-cell

The B lymphocyte, also known as the B-cell is an important component in the immune system. On the surface

of each lymphatic cell are receptors that enable them to recognize foreign substances (antigens). These receptors are very specialized; each can match only one specific antigen. Each type of B-cell has a distinct molecular structure and secretes certain types of Y-shaped antibodies from its surface. The antibody can attach itself to the antigen that it recognizes, leading to its eventual demise. The region on an antibody that is used to recognize and grab the antigen is called *paratope* and the region on any molecular (antigen or antibody) that is attached and recognized by a *paratope* is called *epitope*. (See Fig. 1)

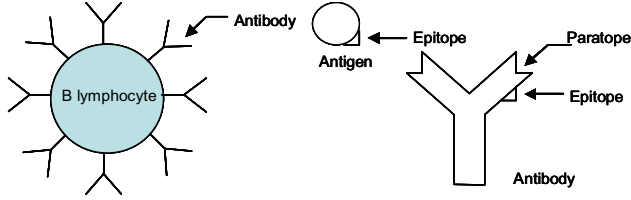


Fig. 1. B-cell, antibody and antigen

C. Clone Selection

Each type of B-cell is genetically programmed to secrete antibodies that are able to recognize one particular antigen. Hence, the immune system as a whole can recognize large numbers of antigens; any type of B-cell is just a tiny proportion of the whole.

When one type of antigen binds to an antibody that recognizes it, the type of B-cell that secretes such antibodies is induced to proliferate rapidly. In other words, it is stimulated. Hence more appropriate antibodies are generated to deal with the infection. One type of antigen may be recognized by several types of antibodies, among which the most suitable are stimulated strongly, while the less suitable receive less stimulation. (See Fig. 2)

D. Negative Selection

The natural immune system is capable of distinguishing virtually any foreign cell from the body's own cells. This is known as self-nonself discrimination [11]. Lymphatic cell have receptors on their surfaces that can detect foreign proteins, known as antigens. However, these receptors are made by random genetic combinations, making it possible to generate some self-reactive receptors. Such self-reactive cells are detected in the thymus. (See Fig. 3) Only those cells that fail to bind to self-proteins are allowed to leave the thymus, entering their stage actively.

III. IMMUNE NETWORK

The *idiotopic network theory* was proposed by Jerne [12] [13]. It indicates that, antibodies, like other molecules, have epitopes. Hence like antigens, they are capable of being recognized by other antibodies. An antibody type is thought to be suppressed when it is recognized, and to be stimulated when it recognizes other types. There is a balance established between recognizing and being recognized.

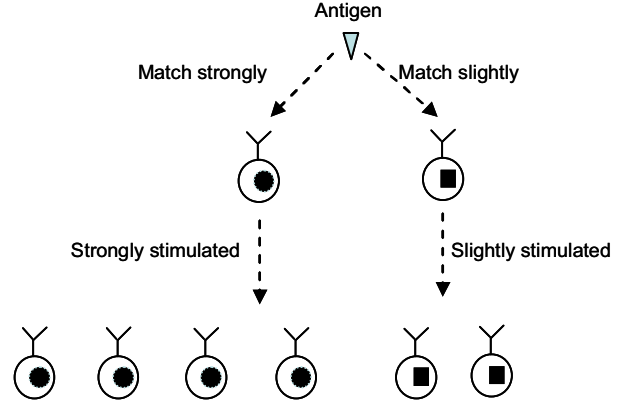


Fig. 2. Clone selection

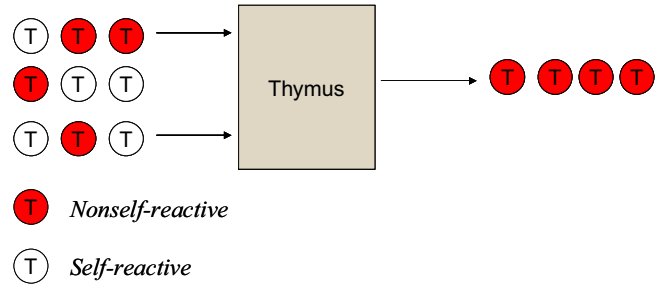


Fig. 3. Negative selection

Based on this theory, Farmer et al. [14] proposed a mathematical model for the immune system. Although it is somewhat simplified, it still makes great sense.

$$\dot{x}_i = c \left[\sum_{j=1}^N M_{ij} x_i x_j - k_1 \sum_{j=1}^N M_{ji} x_j x_i + \sum_{j=1}^n m_{ij} x_i y_j \right] - k_2 x_i \quad (1)$$

$c > 0, M_{ij} > 0, m_{ij} > 0, k_1 > 0, k_2 > 0$

where N is the number of antibody types, $\{x_1, \dots, x_N\}$ are the corresponding concentrations, n is the number of antigen types, $\{y_1, \dots, y_n\}$ are the corresponding concentrations. The first and third terms indicate the stimulation of antibody i recognizing other antibodies and antigens. The second term represents the suppression of antibody i being recognized by other antibodies. Moreover, c is a response rate constant, k_1 represents a possible inequality between stimulation and suppression, and k_2 is the rate of natural death.

IV. SWARM STRATEGY FOR MULTIPLE COMBAT UGVs

The Unmanned Ground Vehicle (UGV) has great potential in both civilian and military applications. The challenge is how to make multiple UGVs cooperate automatically to finish a given task. There is little literature on the subject of Immunology-derived Distributed Autonomous Robotics Architecture (IDARA) [15] [16], where each agent is a model analogous to a B-cell, and the tasks are analogous to the antigens.

In this paper, an algorithm for the Multiple Combat UGV control system is developed using immune system inspired approach. Table 1 shows the analogy between the MCUS and the Immune System (IS). Strictly speaking, an individual agent is not like a B-cell, which carries only one type of antibody. Rather it is like a B-cell container, where several strategies are included.

TABLE I
ANALOGY BETWEEN MCUS AND IS

MCUS	IS
Task	Antigen
Strategy	Antibody
Strategy Weaken	Antibody Suppressed
Strategy Strengthen	Antibody Stimulated
Each agent	B-cell

A. Inspirations of Memory-Based Control Approach

The immune system analogy, as outlined, provides an inspiration for developing a cooperative strategy for the operation of multiple UGVs. The objective of the immune system is to detect and eliminate invasive antigens. Similarly, the objective of the cooperation strategy is to coordinate multiple UGVs to fight against intruding enemies. Combat UGVs are assumed to patrol a certain area and to eliminate intruders wherever they are detected. Once an intruder is detected, the UGV which detects it will attack it and call for nearby UGVs to move closer to its own location. After the intruders are eliminated, the combat UGVs return to roaming as before. In Fig. 4, the top leftmost UGV detects an intruder, thus it attacks and simultaneously communicates to nearby UGVs that it has detected an intruder, and relaying its own position. The closest two UGVs are shown to receive the signal, and to acknowledge the received information. The three UGVs on the right do not detect the intruder and have not received the communication from the first agent; hence they keep roaming and also inform each other to enforce the strategy. The UGV in the middle receives signals from roaming UGVs as well as from the UGV which detects the enemy; hence it will subsequently compare its preference of different strategies, after which it will follow its strongest preference.

B. Mathematical Model

In this section, a mathematical model depicting the information interaction among agents is given as:

$$\dot{S}_i^k(t) = \left(\frac{\alpha}{N} \sum_j \sum_{l \in I^C(k)} m_{ij} c_j^l(t) + \beta \sum_{l \in I^D(k)} \phi_{i,l}^k - \mu \right) c_i^k(t) \quad (2)$$

$$c_i^k(t) = \frac{1}{1 + e^{-\sigma S_i^k(t)}}$$

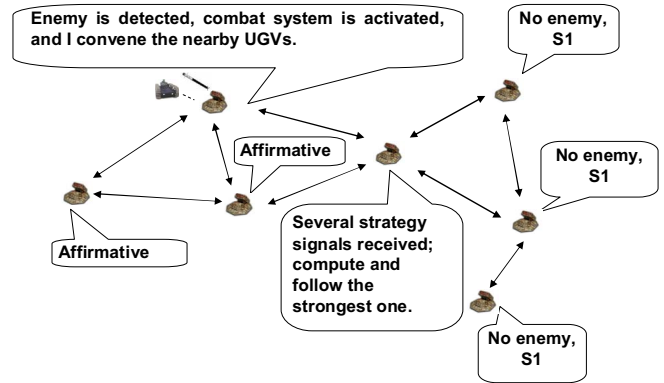


Fig. 4. Illustration of UGV swarm strategy decision

where $S_i^k(t)$ is the strength of strategy i onboard agent k , α and β are “mass” coefficients, $\phi_{i,l}^k$ represents the environmental stimulation of strategy i onboard k , m_{ij} is the mutual interaction coefficient between strategies i and j onboard any two agents, μ is the natural death rate, N is the number of strategies, $I^D(k)$ is the set of all intruders in the detection range of k , and $I^C(k)$ is the set of agents in the communication range of k . $c_i^k(t)$ is a squashed version of $S_i^k(t)$, and it belongs to $[\frac{1}{1+e}, 1]$.

C. Strategies for Each Individual UGV

The strategies onboard each agent are assumed to be of the following types:

- Roaming: denoted R: Dominated by this strategy, The UGV searches the field at random.
- Reporting a detected intruder: denoted D: It is stimulated when the UGV detects an enemy, it convenes the nearby UGVs.
- Following UGV q : denoted F_q : Under this strategy, the UGV heads for the position of UGV q .

D. Decision Making Algorithm

For each UGV, the decision-making algorithm is:

- Step 1: The concentrations of all its strategies are initialized with roaming being the dominant strategy.
- Step 2: The strongest strategy is selected and executed.
- Step 3: After a certain time interval, the UGV reevaluates all its antibody concentrations using local communication.
- Step 4: $t = t + 1$, go to Step 2.

V. SIMULATION VERIFICATION

The simulation was conducted with 16 UGVs in the field. Their initial positions are shown in Fig. 5. The task was to generate the cooperative strategies to command each

vehicle so that they operated cooperatively to detect and eliminate intruders.

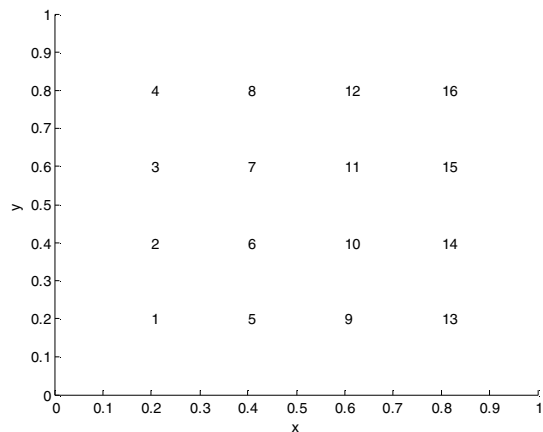


Fig. 5. Initial positions of UGVs

The simulation was implemented in Matlab. There are totally 16 UGVs in the field, which is a 1×1 unit square, the detection radius: 0.1 and the communication radius: 0.15, $\alpha = 0.5$ and $\beta = 1$, and m_{ij} as defined in Table 2 (If $p = q$, $c = 1$, otherwise $c = -0.4$)

The UGVs roamed at the beginning and the result is shown in Fig. 6. Next, an enemy, denoted by a red star, appeared at location (0.15, 0.15). The nearby UGVs detected it and headed for it, while the distant ones kept roaming. Fig. 7 shows the effect of the proposed algorithm after 10 iterations.

TABLE II
STRATEGY MATCHING SPECIFICATIONS

i	R	R	R	D	D	D	F_q	F_q	F_q
j	R	D	F_q	R	D	F_q	R	D	F_q
m_{ij}	1	-0.8	-0.4	-0.4	1	-0.4	-0.6	-0.4	c
m_{ji}	1	-0.4	-0.6	-0.8	1	-0.4	-0.4	-0.4	c

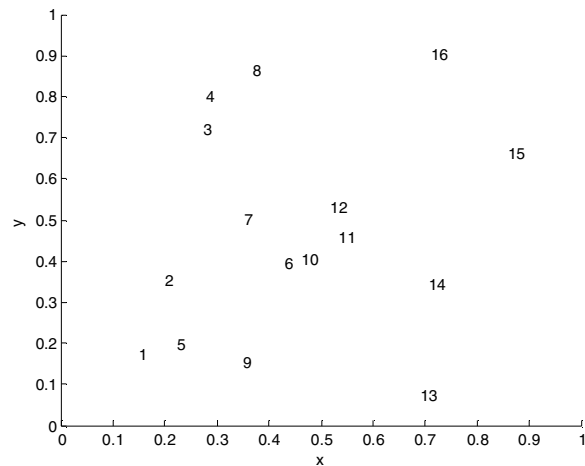


Fig. 6. Effect of roaming after 10 iterations

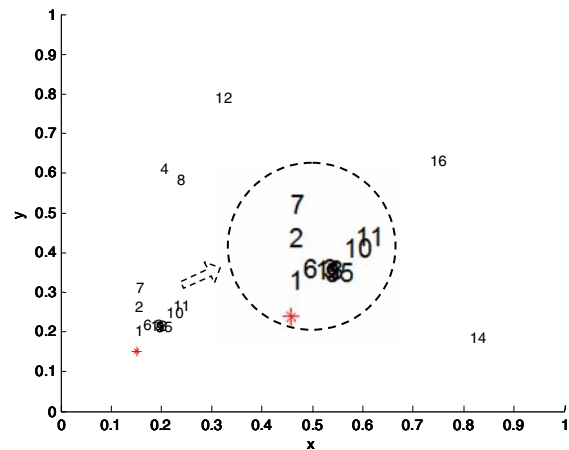


Fig. 7. Effect of one intruder detected after 10 iterations

Three intruders were then introduced to the field simultaneously and immediately after 10 iterations of roaming. The intruders were assumed fixed at positions (0.15, 0.15) (0.35, 0.35) (0.85, 0.85). The proposed algorithm worked reasonably well. Note that each intruder is attacked by several UGVs, and there are still some UGVs roaming free. Fig. 8 shows the effect after 10 iterations.

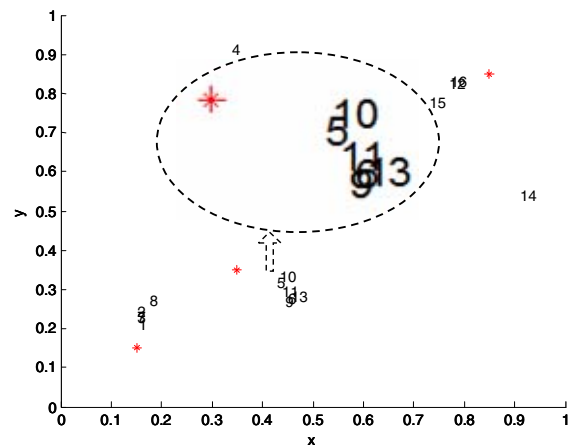


Fig. 8. Effect of multiple intruders detected after 10 iterations

A. Discussion

In this section, the performance of the algorithm is evaluated with different detection and communication radius.

- In case 1, the detection radius was chosen as 0.1 and the communication radius was chosen as 0.5. The result is shown in Fig. 9.
- In case 2, the detection radius was chosen is 0.5 and the communication radius was chosen as 0.1. The result is illustrated in Fig. 10.
- In case 3, both the detection radius and the communication radius were chosen as 1. The result is depicted in Fig. 11.

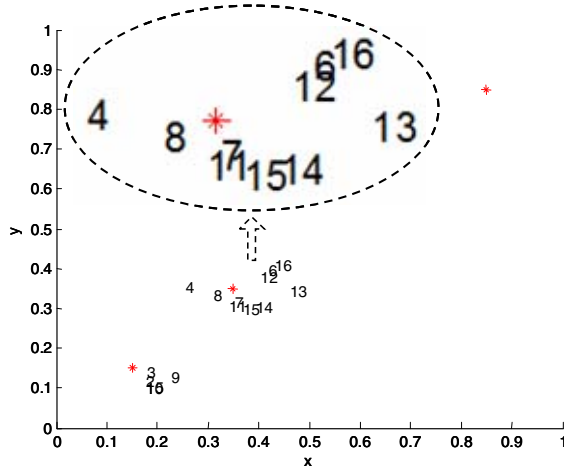


Fig. 9. Snapshot after 10 iterations of case 1

From these simulation results, it is observed that with a wider detection radius or communication radius, the agents are more likely to group together (less likely to have free roaming agents), and that with an extreme detection radius and an extreme communication radius, the UGVs become confused by a sensory overload. They no longer attack the intruders effectively.

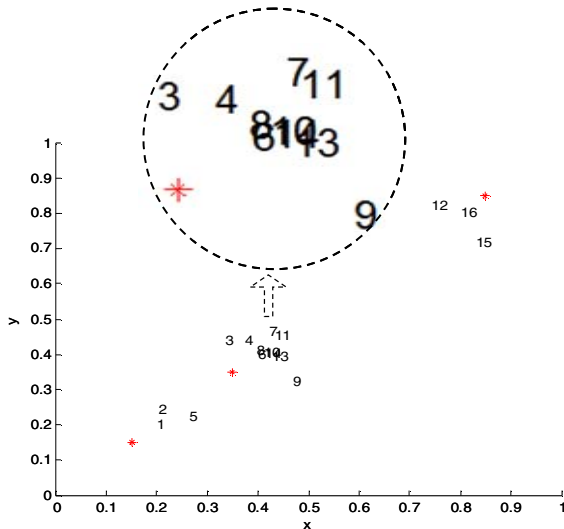


Fig. 10. Snapshot after 10 iterations of case 2

B. Introduction of a Critic

The simulation results show that agents are allergic to some high task density areas and are crowded in that region. By analysis, the problem is due to strategy exploration, which means that too many agents are performing the same strategy. In order to overcome this deficiency, a “critic”, which is inspired by the negative selection function of the thymus, is introduced to the system. The critic can limit the number of agents which are conducting the same strategy through negative selection. Simulation results are presented as follows:

- In case 1, the detection radius was chosen as 0.1 and the communication radius was chosen as 0.5. The result is presented in Fig. 12.
- In case 2, the detection radius was chosen as 0.15 and the communication radius was chosen as 1. The result is plotted in Fig. 13.
- In case 3, both the detection radius and the communication radius were chosen as 1. The result is presented in Fig. 14

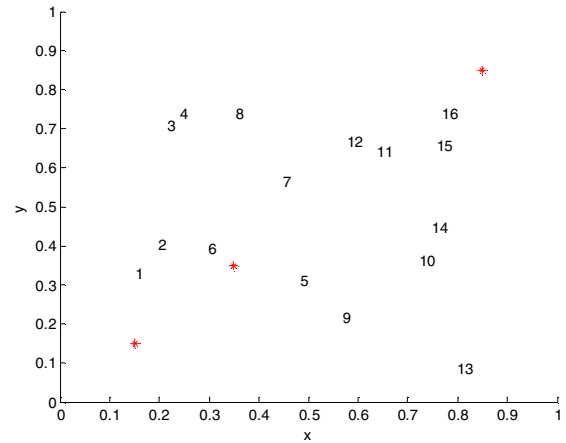


Fig. 11. Snapshot after 10 iterations of case 3

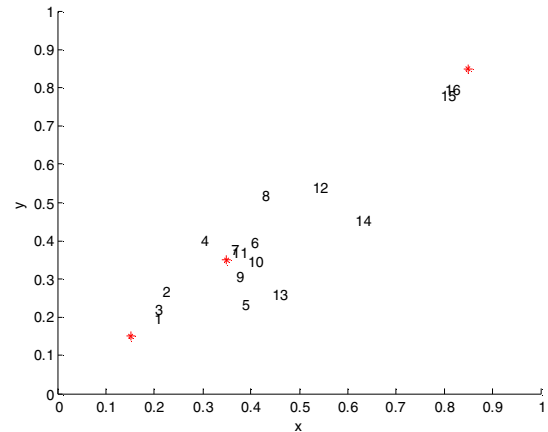


Fig. 12. Snapshot after 10 iterations of case 1 (Thymus included)

In Fig. 12 and 13, the swarm coordination is clearly much improved, even when the communication radius is 1. Agents can work more intelligently and cooperatively on three targets and are no longer as crowded as when there was no critic. However, when the detection ratio approaches 1, the agents appear to be confused and begin purposeless roaming without attending to the intruders. The problem can be simply resolved by assigning different priorities to the targets. In Fig. 12, agents 12 and 14 are roaming, but they are also strategically located between two enemies. This shows the great promise of the proposed algorithm. However, when detection radius goes too large, even with the help of the critic, the system is not likely to work well, as shown in Fig. 14. Because the agent detects several targets

simultaneously, it does not know where to go. This issue should be considered in future works.

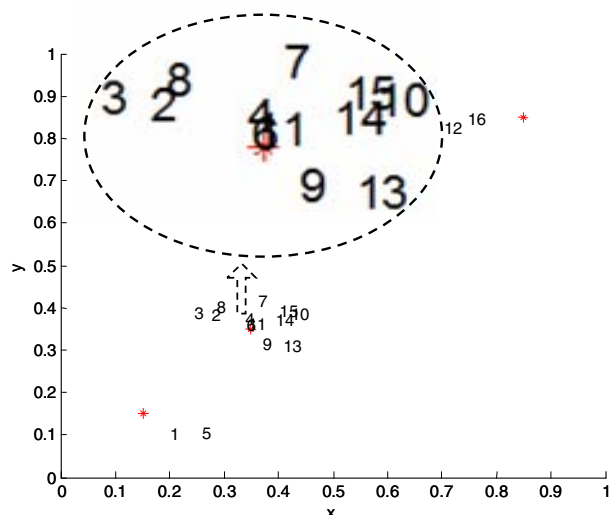


Fig. 13. Snapshot after 10 iterations of case 2 (Thymus included)

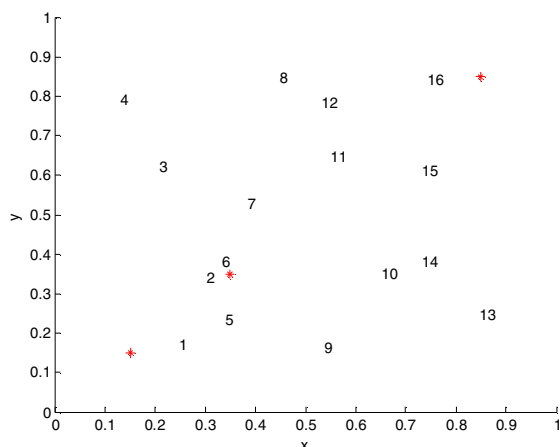


Fig. 14. Snapshot after 20 iterations of case 3 (Thymus included)

VI. CONCLUSION

Immune system based swarm strategy algorithms are proposed in this paper. As shown by simulation, this method works satisfactorily in general. However, if the communication ratio is increased, the agents tend to crowd around the target which is first detected by most agents. This sensitivity can be understood intuitively but is undesirable. In order to overcome this deficiency, a “critic” whose operation is inspired by the thymus is introduced. The critic limits the number of agents which conduct the same strategy and require very limited global information. Simulation shows that the critic can significantly reduce the strategy explosion and avoid agent crowding. The swarm strategy algorithm is efficient, converges quickly, and the intruders, in most cases, are detected if the parameters are set properly.

Because of the distributed and decentralized nature, bio-inspired systems are especially efficient and superior when

they are applied to very large-scale agent systems. Hence the computation complexity does not grow exponentially with the number of the agents. Although some robustness has been observed, issues relating to robust and adaptive capabilities must be further studied to make this method more feasible and practical.

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